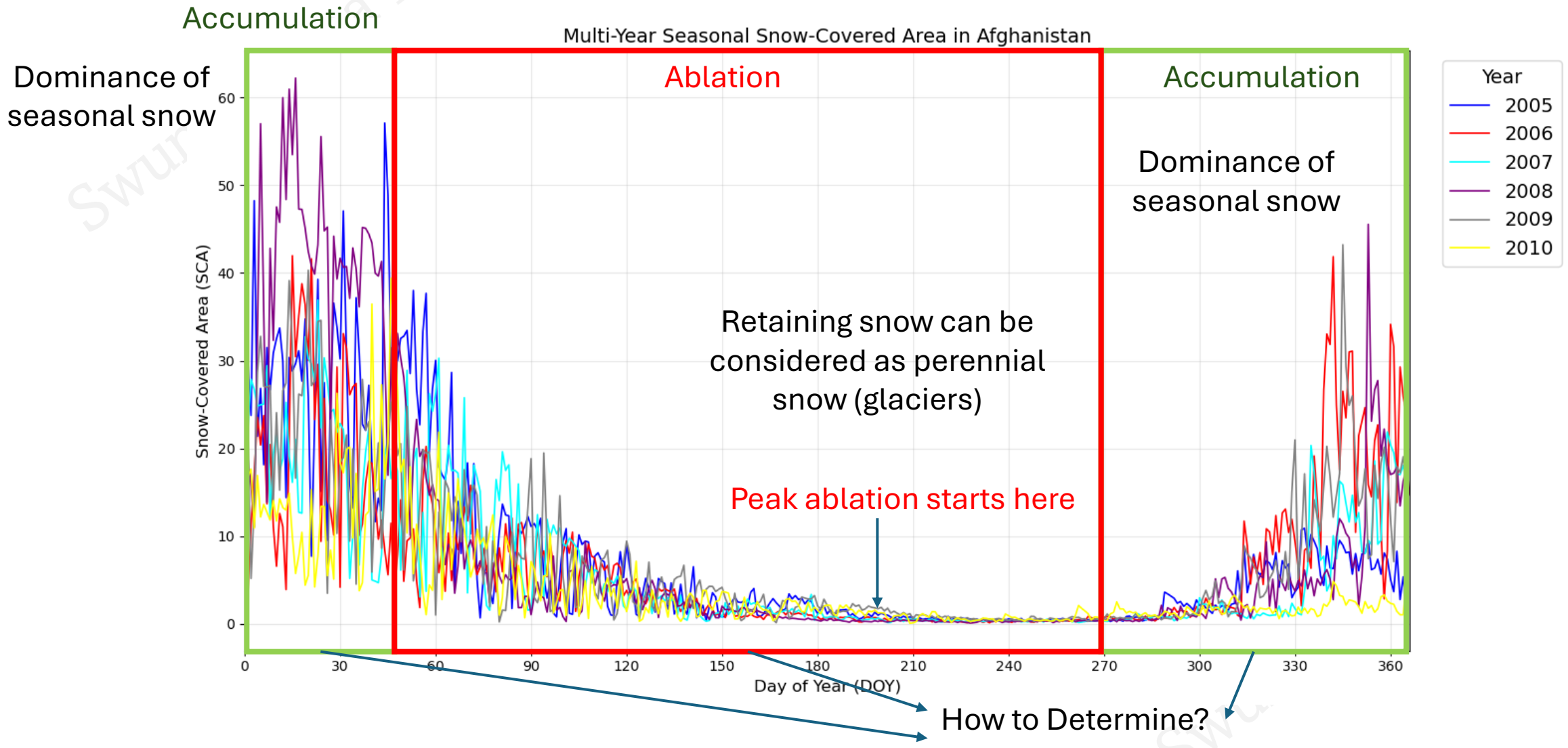
Date Range Determination of Seasonal Snow-Free Glaciers Using MODIS Snow Products

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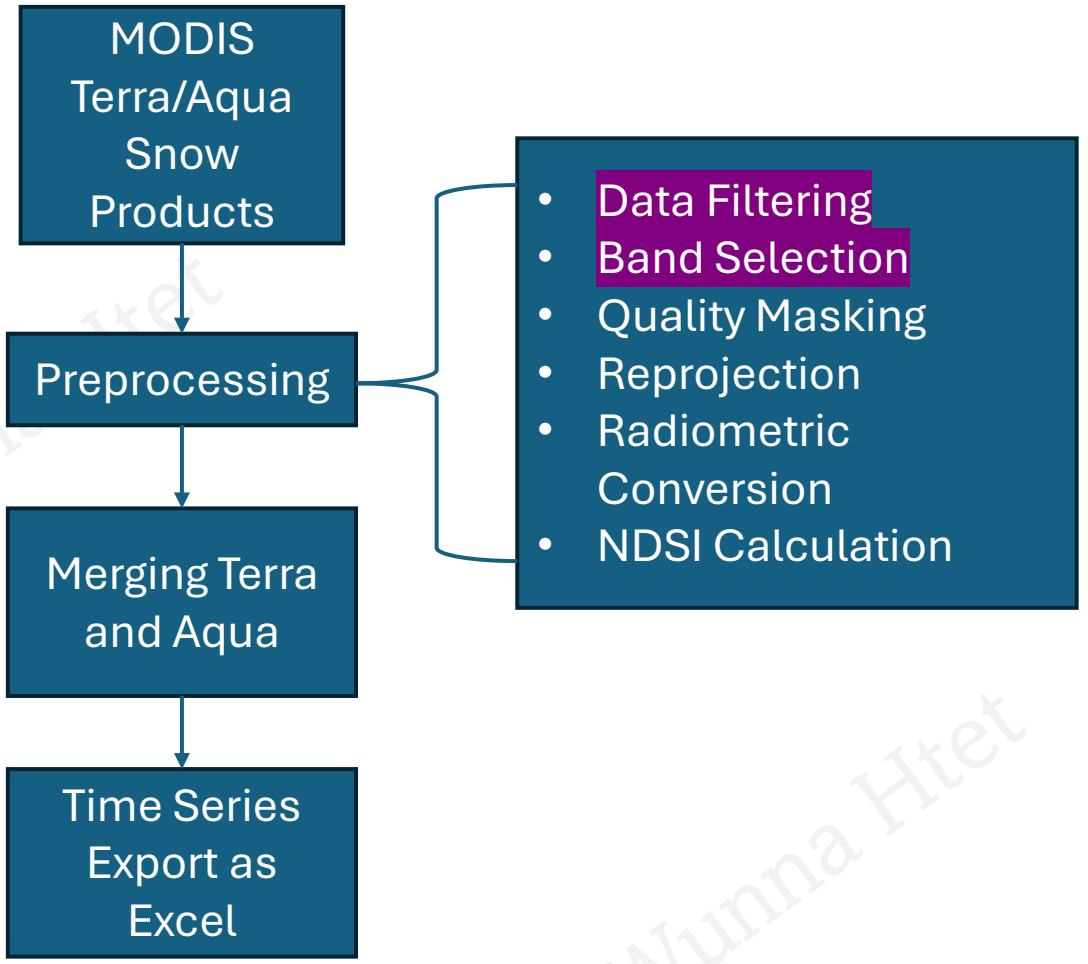
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Seasonal snow obstruct presence of perennial glaciers



Overall conduct

- **Main task**:
  - To extract monthly time series of snow-covered area SCA from 2005 to 2010 as csv file.
- **Dataset:** MODIS Terra/Aqua Snow Cover Daily Global 500 m
- Check out how to visualize time series data in python script.



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Uploading MODIS snow datasets

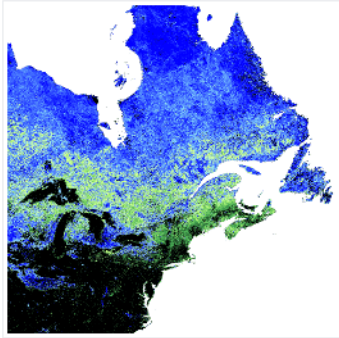
Earth Engine Data Catalog

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Home | Categories | All Datasets | All Tags | Landsat | MODIS | Sentinel | Publisher | Community | API Docs | Dataset Status | Changelog

MOD10A1.061 Terra Snow Cover Daily Global 500m

Page Summary



Dataset Availability

2000-02-24T00:00:00Z–2026-09-11T00:00:00Z

Dataset Producer

NASA NSIDC DAAC at CIRES

Earth Engine Snippet

```
ee.ImageCollection("MODIS/061/MOD10A1")
```

Cadence

1 Day

Tags

albedo | cryosphere | daily | geophysical | global | modis | nasa | nsidc | snow | terra | mod10a1

Description | Bands | Terms of Use | Citations | DOIs

Bands

Pixel size: 500 meters (all bands)

Name	Min	Max	Scale	Pixel Size	Description
NDSI_Snow_Cover	0	100		500 meters	NDSI snow cover. This value is computed for MOD10_L2 and retrieved when the observation of the day is selected. Provider values above 100 are masked out in this band (they can be found in the "NDSI_Snow_Cover_Class" band).
NDSI_Snow_Cover_Basic_QA				500 meters	A basic estimate of the quality of the algorithm result. This value is computed for MOD10_L2 and retrieved with the corresponding observation of the day.

Imports (3 entries)

```
var aoi: Table projects/ee-swunait/assets/fao_afg_glaciers/Afghanistan_boundary_v1
var terra_snow: ImageCollection MODIS/061/MOD10A1
var aqua_snow: ImageCollection MODIS/061/MYD10A1
```

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Filtering Dataset

```
////////////////////////////////////////////////////////////////////////
// DEFINE YOUR SET ANALYSIS PERIOD
////////////////////////////////////////////////////////////////////////

var startyear = 2005, endyear = 2010;

var start = ee.Date(startyear+'-01-01'), end = ee.Date((endyear+1)+'-01-01');

////////////////////////////////////////////////////////////////////////
// FILTER DATASETS
////////////////////////////////////////////////////////////////////////


terra_snow = terra_snow.filterDate(start, end).filterBounds(aoi);
aqua_snow = aqua_snow.filterDate(start, end).filterBounds(aoi);
```

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Quality Masking using Quality Band Bits

```
function PREPROCESS(img, bandname) {

  // Reproject and clip
  var reproj = img.reproject({
    crs: 'EPSG:4326',
    scale: 500
  }).clip(aoi);

  // -------------------------
  // Basic QA
  // -------------------------
  var qr = reproj.select('NDSI_Snow_Cover_Basic_QA');

  // Extract the relevant QA bits (bits 0-1)
  var qa = qr.bitwiseAnd(3);

  // Keep Best (0), Good (1), and OK (2)
  var qaMask = qa.lte(2);

  // -------------------------
  // Valid snow-cover values
  // -------------------------
  var snowMask = reproj.select('NDSI_Snow_Cover').lte(100);

  // -------------------------
  // Apply masks
  // -------------------------
  reproj = reproj.updateMask(qaMask).updateMask(snowMask);

  // -------------------------
  // Select snow-cover band
  // -------------------------
  reproj = reproj.select('NDSI_Snow_Cover').rename(bandname);

  // -------------------------
  // Date
  // -------------------------
  var date = ee.Date(img.get('system:time_start')).format('YYYY-MM-dd');

  return ee.Image(reproj.copyProperties(img, ['system:time_start']).set('Date', date)
  );
}
```

Reprojection

Quality Masking

```
////////////////////////////////////////////////////////////////////////
// PREPROCESSING OF QUALITY MASKED MODIS SNOW DATASET
////////////////////////////////////////////////////////////////////////

terra_snow = terra_snow.map(function(img) {
  return PREPROCESS(img, 'Snow_Terra');
});

aqua_snow = aqua_snow.map(function(img) {
  return PREPROCESS(img, 'Snow_Aqua');
});
```

Let's say you have pixel bit value of **40** (number).

Bit: 5 4 3 2 1 0

↓ ↓ ↓ ↓ ↓ ↓

Binary: 1 0 1 0 0 0

Bit definition

Bit 0 = Fill (1-yes/0-no)
Bit 1= Cloud (1-yes/0-no)
Bit 3 = Snow (1-yes/0-no)
Bit 5 = Cirrus (1-yes/0-no)

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Merging both Terra and Aqua based on their date

```
//////////////////////////////////////////////////////////////////////
// 8. MERGE TERRA AND AQUA SNOW DATASETS
//////////////////////////////////////////////////////////////////////

var snow_combined = terra_snow.merge(aqua_snow);
```

Merging both image collections

```
//////////////////////////////////////////////////////////////////////
// 9. COMBINE TERRA AND AQUA BASED ON ACQUISITION DATE
//////////////////////////////////////////////////////////////////////

// Get unique dates
var snow_dates = ee.List(
  snow_combined.aggregate_array('Date')
).distinct().sort();
```

Collect all unique dates from both collections

```
// Create one image for each day
var snow_daily = ee.ImageCollection.fromImages(
  snow_dates.map(function(date) {

    date = ee.String(date);

    // Get all Terra + Aqua images acquired on this date
    var daily = snow_combined
      .filter(ee.Filter.eq('Date', date))
      .median()
      .rename('snow_cover');

    return daily
      .set('Date', date)
      .set(
        'system:time_start',
        ee.Date(date).millis()
      );
  })
);
```

Find images of the same date and apply median operation

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Create time series data of SCA

```
////////////////////////////////////////////////////////////////////////
// CREATING A TABLE OF EXTRACTED ZONAL VALUES OF SCA
////////////////////////////////////////////////////////////////////////
var table = ee.FeatureCollection(

  dateList.map(function(date) {

    date = ee.String(date);

    ////////////////////////////////////////////////////////////////////
    // SNOW COVER
    ////////////////////////////////////////////////////////////////////

    var snow_image = snow_daily
      .filter(ee.Filter.eq('Date', date))
      .first();

    var snow_value = ee.Algorithms.If(
      snow_image,
      ee.Image(snow_image)
        .reduceRegion({
          reducer: ee.Reducer.mean(),
          geometry: aoi,
          scale: 500,
          maxPixels: 1e13
        })
        .get('snow_cover'),
      null
    );

    ////////////////////////////////////////////////////////////////////
    // CREATE OUTPUT FEATURE
    ////////////////////////////////////////////////////////////////////

    return ee.Feature(null, {
      date: date,
      snow_cover: snow_value
    });
  })
);
```

For each date of image

Extract zonal mean value of Snow Covered Area (SCA)

Important: the scale plays an essential role in computation cost.

Return a row with two properties: date and SCA

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Finally, export out your time series SCA of MODIS

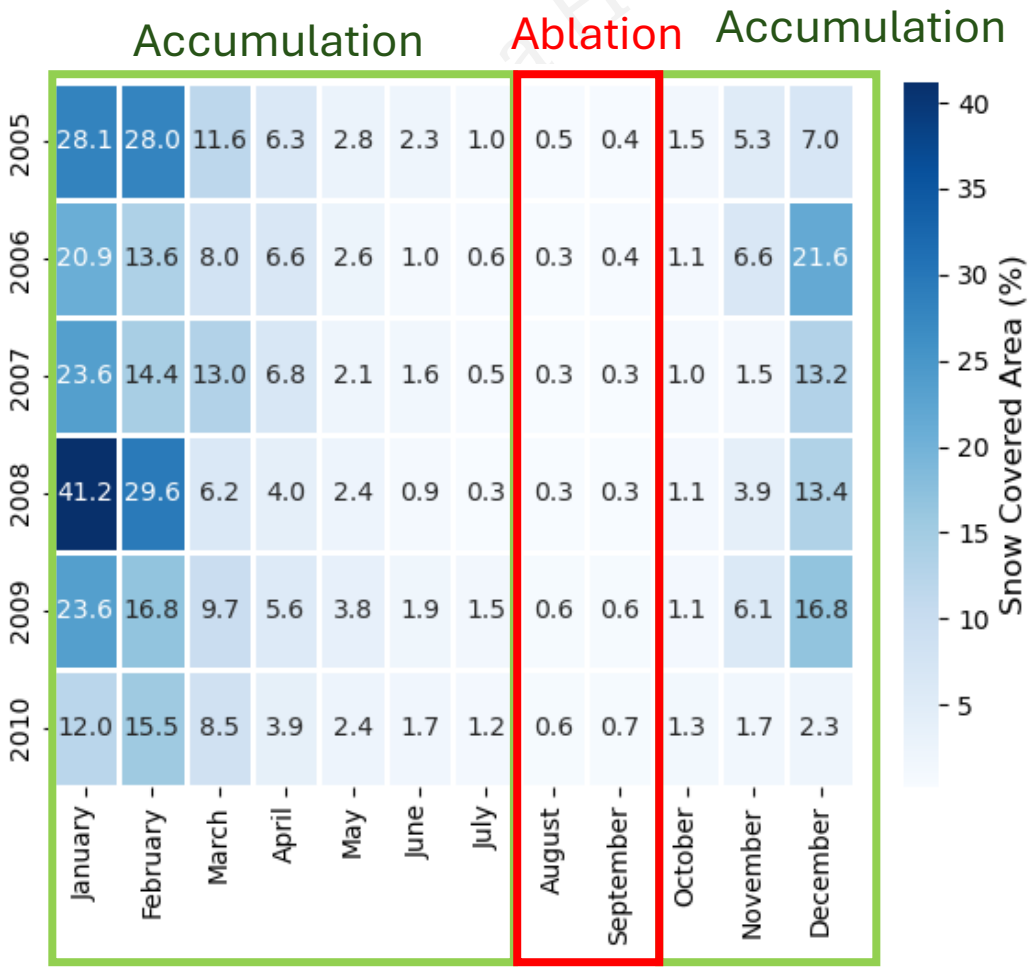
```
////////////////////////////////////////////////////////////////////////
// EXPORT CSV TO GOOGLE DRIVE
////////////////////////////////////////////////////////////////////////

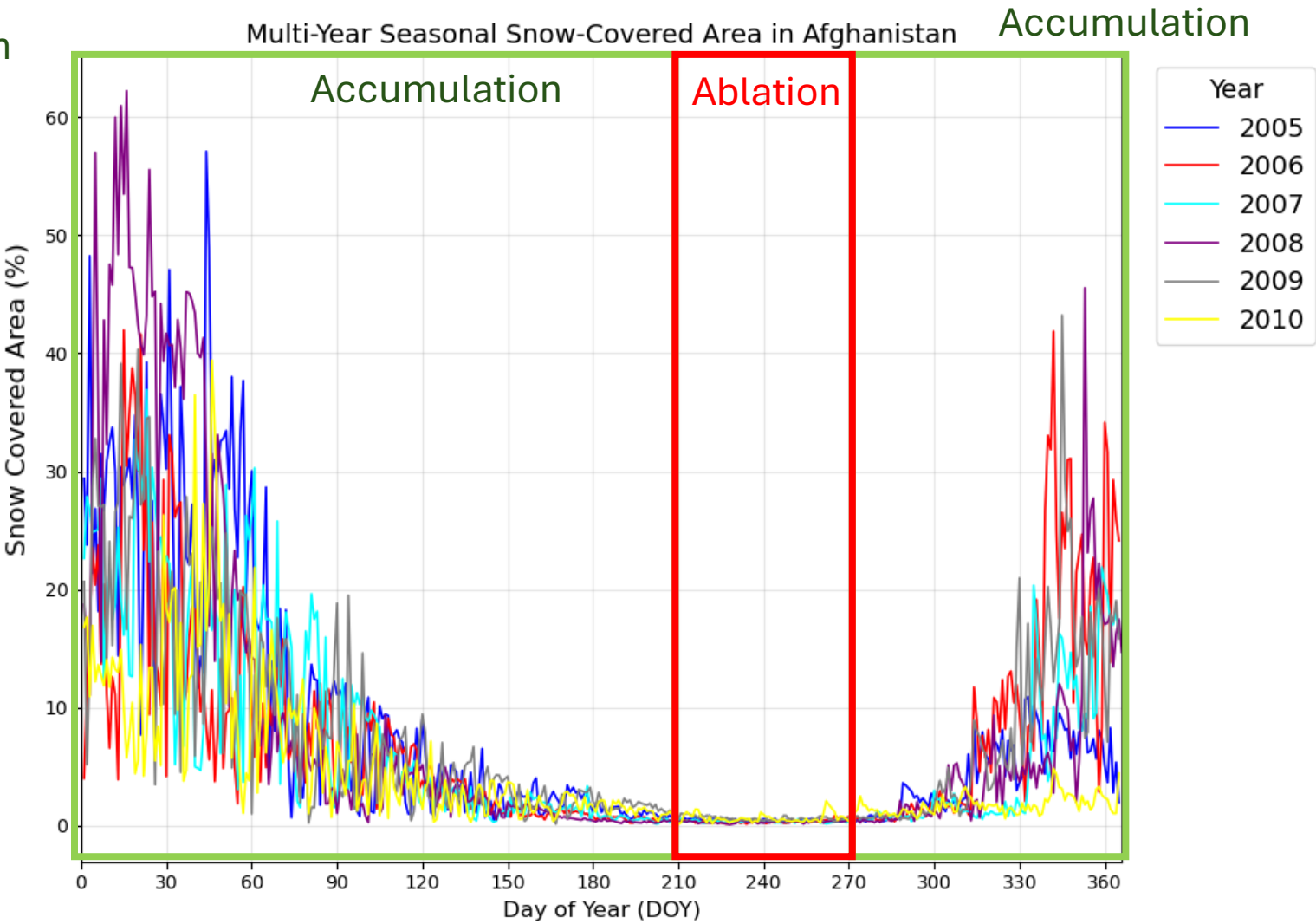
Export.table.toDrive({

  collection: table,

  description:'Afghanistan_SCA_TimeSeries_' + startyear + '_' + endyear, // Change here
  folder:'FAO Afghanistan Glaciers Project', // Change here
  fileNamePrefix:'Afghanistan_LST_Snow_CHIRPS_TimeSeries_' +startyear + '_' + endyear,// Change here
  fileFormat: 'CSV',
  selectors: ['date','snow_cover']
});
```

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***Concluded**: August-September window is the ideal time range for filtering satellite imagery for glaciers with the least seasonal snow extents.*